Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/23/2015
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Project Title:	Eutectic Solidification in a Ternary Alloy		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
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	Note this is a follow-on project for Dr. Ralph Napolitano's project,"Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy," which ended in 2014. Solidification processing of metallic alloys can produce a wide array of growth morphologies, which can be manipulated to access new structures, materials, and functionalities. The array of accessible structures is particularly wide in ternary and other higher-order systems, but the challenge of controlling the many variables that influence growth structures is immense and requires a better understanding of the interplay between the mechanisms of pattern formation and the influence of applied forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal		

Task Description:	field. The two principal objectives to be addressed are (i) to establish an analytical framework for 2D and 3D quantitative description of multiphase microstructures, and (ii) to employ the newly developed quantitative framework to investigate the dynamics of pattern formation in ternary eutectic alloys. The work will focus primarily on the invariant three-phase eutectic structure. Ground-based samples will be processed via a vertical Bridgman-style directional solidification furnace. Electron backscatter diffraction will be a major characterization tool to measure crystallographic orientation, texture, and misorientation relationships. Analysis over the length of the sample will provide insight into initial transient behavior, grain competition, and steady state growth. The relationship between geometric and crystallographic parameters, such as curvatures, orientations, and periodic ordering, will be used to quantify the morphological trajectory of the eutectic and identify fundamental principles of pattern formation. Seeded directional solidification (DS) experiments will be conducted to compare the dynamics associated with specific crystal orientations. In order to validate the experimental correlations and analytical models developed from the findings of the ground-based work, carefully selected flight-based directional solidification experiments will be conducted using the European Space Agency (ESA) Materials Science Lab (MSL) and the Low Gradient Furnace (LGF). The project will continue to provide collaborative support to the European Space Agency's project titled "Solidification along a Eutectic Path in Ternary Alloys" (SETA), which has already accomplished important work using the Al-Ag-Cu alloy system. Significant findings of the SETA project to date have been in the areas of pattern development and the role of fault structure for both univariant and invariant eutectic compositions, the role of crystal orientation and "locked" grains, and development of nonplanar growth modes. Facilitating open scientific interaction between the PI and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. The planning and performance of flight-based solidification experiments is being done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As mentioned in the Task Description, the objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is far-reaching, given the extremely broad range of multiphase structures that can be fabricated through solidification processing, and is certainly not limited to metals and alloys. Indeed the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
	The technical progress achieved during the first year of this program is outlined here. Many first-year activities were related to the establishment of experimental practices, the development of dedicated research facilities, and the assembly and training of student research assistants. Specific areas of focus have been (i) design and build of a directional solidification furnace, and (ii) characterization and analysis of the Al-Ag-Cu three-phase invariant eutectic structure after directional solidification. The second task was carried out on four fronts in conjunction with a variety of collaboration partners as described below. DESIGN AND BUILD OF DIRECTIONAL SOLIDIFICATION FURNACE In order to perform controlled, ground-based experiments at University of Alabama at Birmingham (UAB), the Principal Investigator (PI) and four students worked together to design and build a vertical Bridgman-type furnace. A thermal gradient is created inside the furnace chamber by a Kanthal wire resistance heating element supported by a mullite shell and a water-cooled copper chill block. The heating element was selected specifically to support the range of temperatures needed to investigate Al-Ag-Cu (700-1200°C). The chamber is insulted with two types of zirconia insulation. A screw drive and stepper motor control vertical translation of the furnace unit. Samples will be contained in a suspended crucible. A quick release mechanism was designed in order to allow the crucible and sample to drop smoothly into a container of water to accomplish quenching. The furnace sits on a levitated, vibration damping table to minimize external influences on microstructure development during processing. An inert argon atmosphere will be maintained by direct attachment of a tube to the crucible. The furnace was finished in June 2015 and test processing is currently underway. SOLID STATE EVOLUTION OF INVARIANT EUTECTIC MORPHOLOGY Understanding changes that occur during processing of directionally solidified Al-Ag-Cu is an important component of analysis. In this system there is a particularly large decrease in maximum solubility of Ag in the Al phase just below the eutectic temperature (equilibrium value drops from 18% to 8% in 20°C). During directional solidification, especially with slow velocities and low thermal gradients, the microstructure undergoes significant changes immediately after freezing. The true solidification morphology then can only be observed directly behind a quenched solid-liquid interface. Analysis of quenched samples is necessary to understand the changes that occur, and obtain a proper interpretation of these ex-situ microstructures generally. This is a critical study since the findings will influence the interpretation of flight experiments. The PI worked with NASA PI R. Napolitano (Iowa State University (ISU)) and his student I. Sargin (PhD, July 2015) to analyze and publish the results in this section based on experiments conducted at ISU. Transverse cross sections were taken from a ternary eutectic sample at increasing distances behind the quenched interface. An increase in volume of the Ag2Al phase was obvious, along with a change in the contract of the phases when obtained by backscatter SEM, indicating significant composition differences. Samples were analyzed as a function of distance from quench in a variety of ways including phase fraction, composition, phase size, and length of certain boundary segments. Based on these results, a cellular automaton model was developed to evolve the microstructure forward or backwards in position. Application of the model produced a reasonable estimation of the structure at the growth front based on the structure measured at some known distance behind the front. CHARACTERIZATION OF INVARIANT EUTECTIC BY PRINCIPAL COMPONENT ANALYSIS Because the three-phase eutectic microstructure in Al-Ag-Cu is very complex, a single measure like lamellar spacing

Task Progress:	cannot be used to fully quantify the structure. A collaboration with Surya Kalidindi (Georgia Tech (GT)) and his student Yuksel Yabansu was begun to apply their technique for principal component analysis (PCA) via pair correlations to Al-Ag-Cu three-phase eutectic microstructures. The PCA technique is used to show variation in datasets. It automatically emphasizes the factor that contributes most to the variation by plotting it along PC axis 1, the second biggest factor along PC 2, etc. Thus it requires no a priori assumptions about what feature of a dataset (such as a collection of micrographs) is changing. PCA analysis was performed on six sets of experimental data obtained from I. Sargin at ISU. It was shown that the PCA is able to discriminate between the microstructures produced at different velocities. However, which particular factor(s) of the different microstructures are producing these results cannot be easily determined. Therefore, the PI is working with the GT group on a systematic study of idealized microstructures related to those seen in Al-Ag-Cu. COMPARISON OF EXPERIMENTAL TO SIMULATED MICROSTRUCTURES This year the PI began a collaboration with the research group of Britta Nestler at the Karlsruhe Institute of Technology in Karlsruhe, Germany, who is currently doing large-scale phase field modeling of 3D ternary eutectic growth in Al-Ag-Cu. This work did not consider solid-state diffusion, instead simulating a solidification microstructure that matched the evolved morphology discussed above. The PI was able to provide images of quenched samples and will continue to work with the modeling group going forward to correctly simulate the true solidification structure. In addition, a second collaborative paper has been submitted considering the possibility of spiraling behavior in these eutectics, a possibility for which the PI has seen some. Simulations of two-dimensional lamellar growth were used to determine the conditions under which maximum lamellar tilt could be achieved. This tilt was the result of non-mirror symmetric phase boundary arrangements, not interface anisotropy, a condition that had not previously been reported. Application of these same interface conditions in 3D growth produced spiraling between two of the rod-type phases as shown. Future work will attempt to confirm the presence of this phenomenon experimentally. GRAIN COMPETITION AND CRYSTALLOGRAPHIC ORIENTATION Crystallographic orientation and the orientation relationships (OR) between phases play a major role in the solidification behavior of complex eutectics. In order to investigate this, electron backscatter diffraction (EBSD) was used to analyze the crystal orientation in different three-phase eutectic grains and to find the common crystallographic planes between phases. Several unique crystallographic ORs have been identified. Current work is focused on grain competition and the persistence or disappearance of particular orientations over the growth distance, as well as the correlation between crystal orientation and morphology. Because a special ion polishing technique is necessary to obtain EBSD patterns from all three phases, and this type of polisher is relatively rare, the PI applied for and received time at the Oak Ridge National Lab microscopy user center, where some work was done in February 2015. Additional EBSD analysis was carried out in June using the facilities at SETA project collaborator Access in Aachen, Germany. Notably, EBSD was carried out on several sections at different solidification distances for a sample where the solidification velocity was increased. A change in morphology was observed as velocity increased, but was found not to correlate with a change in crystal orientation. The adaption of morphology and phase spacing in response to velocity changes, including possible branching mechanisms, is a subject of current study.
Bibliography Type:	Description: (Last Updated: 06/05/2023)
Articles in Peer-reviewed Journals	Hotzera J, Jainta M, Steinmetz P, Nestler B, Dennstedt A, Genau A, Bauer M, Kostler H, Rude U. "Large scale phase-field simulations of directional ternary eutectic solidification." Acta Materialia. 2015 Jul;93:194-204. http://dx.doi.org/10.1016/j.actamat.2015.03.051 , Jul-2015